

Identifying High-Risk Patients for 30-Day Hospital Readmission Using Healthcare Data

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Abstract—Hospital readmission within 30 days is a critical challenge for global healthcare systems, impacting patient care quality and increasing institutional costs. This study addresses the scalability gap in existing literature, which often relies on small-scale datasets and traditional analytical tools, by implementing a distributed predictive framework using PySpark on 30,000 healthcare records. Aligned with Sustainable Development Goal 3 (SDG 3), this research leverages Apache Spark’s computing capabilities to identify high-risk patients based on clinical and demographic variables. Exploratory data analysis (EDA) revealed that chronic conditions are major risk factors, where patients with only diabetes (13.3%) and those with both diabetes and hypertension (13.1%) exhibit the highest readmission risks. Additionally, patients discharged to rehabilitation centers were identified as the highest-risk group with a 17.5% readmission rate. Experimental results demonstrate that Logistic Regression and Random Forest achieved superior performance, both delivering an F1-score of 0.8187. These findings establish a scalable infrastructure for proactive clinical interventions, supporting healthcare providers in optimizing resources and enhancing patient outcomes through big data analytics.

Index Terms—Apache Spark, big data analytics, healthcare informatics, hospital readmission, machine learning, PySpark, scalability, Sustainable Development Goals (SDG 3).

I. INTRODUCTION

Hospital readmission within 30 days is a common problem in healthcare systems worldwide. High readmission rates increase hospital workload and healthcare costs and may indicate gaps in patient care or ineffective discharge planning. Identifying patients who are at high risk of readmission is an important step toward improving healthcare quality and reducing unnecessary hospital visits. By analyzing healthcare data, it becomes possible to detect patterns and factors associated with patient readmission, which can support better healthcare decisions and improve patient outcomes.

This project aligns with Goal 3 (SDG 3): Good Health and Well-Being, which focuses on improving healthcare quality

and promoting better health outcomes. Reducing hospital readmission contributes to enhancing healthcare efficiency and ensuring better patient care.

The dataset used in this study contains 30,000 patient records with 12 variables related to patient demographics, medical conditions, and hospitalization details. These variables provide important information that may influence whether a patient will be readmitted to the hospital within 30 days after discharge. The dataset includes features such as `patient_id`, `age`, `gender`, `blood_pressure` (mmHg), `cholesterol` (mg/dL), `BMI` (kg/m²), `diabetes`, `hypertension`, `medication_count`, `length_of_stay`, and `discharge_destination`.

The dataset provides valuable insights into patient health conditions and hospitalization patterns. These features enable the identification of key risk factors associated with hospital readmission, such as chronic diseases, length of stay, and medication usage. This supports more informed clinical decision-making and improves patient outcomes.

The target variable in the dataset is `readmitted_30_days`. It indicates whether the patient was readmitted to the hospital within 30 days after discharge. Analyzing these variables helps identify patterns associated with hospital readmission and supports the development of predictive models.

Therefore, the main research question addressed in this study is: How effectively can machine learning techniques be used to predict 30-day hospital readmission using healthcare patient data?

II. RELATED WORK

Several studies have explored the use of machine learning techniques to predict hospital readmission and improve healthcare decision-making. With the increasing availability

of healthcare data, various models have been developed to identify high-risk patients and enhance prediction accuracy. These approaches aim to support healthcare providers by enabling early intervention and reducing unnecessary hospital readmissions.

Oh et al. [2] conducted a retrospective study to develop early prediction models for unplanned 30-day readmissions among high-risk patients. The study aimed to enhance the identification of high-risk individuals by integrating nursing data, such as ward severity and fall risk assessments, with traditional clinical and demographic variables. The authors compared multiple machine learning algorithms, including random forest and CatBoost, to evaluate model performance. The results identified body mass index (BMI), systolic blood pressure, and age as significant predictors of readmission risk. This study supports our project as we analyze similar clinical features, including blood pressure and BMI, to develop an effective classification model using a dataset of 30,000 patient records [1].

Alajmani and Elazhary [3] explored the use of machine learning techniques to predict hospital readmission specifically among diabetic patients. The study aimed to identify the most accurate approach for this population by comparing several models, including logistic regression, decision trees, support vector machines (SVM), naïve Bayes, and neural networks. The results showed that the support vector machine achieved the highest performance among the tested algorithms. This study is relevant to our project because it highlights the importance of model comparison, which aligns with our objective to apply and evaluate different machine learning models using PySpark to determine the most effective predictor.

Glans et al. [4] developed and validated a risk assessment tool to identify patients at higher risk of medication-related readmission within 30 days of discharge. The study aimed to improve prediction accuracy by considering medication-related factors alongside previous hospital visits and chronic conditions, particularly among older adults. The findings indicated that incorporating these specific factors significantly enhances the identification of high-risk patients. This study supports our project as our dataset includes similar patient-related features, allowing us to explore these indicators to enhance the overall prediction performance of our model.

Halac et al. [5] developed and validated a machine learning model to predict 30-day hospital readmissions in a healthcare setting. The study aimed to support the early identification of patients who are more likely to be readmitted after discharge. The authors used electronic health record data and compared several machine learning algorithms, including logistic regression, random forest, and LightGBM. The results showed that random forest achieved the best performance compared with the other models. This study supports our project because it uses healthcare data and machine learning techniques to identify high-risk patients and improve hospital readmission prediction.

Rashidi et al. [6] conducted a systematic review of 24 studies to identify key predictors of 30-day readmission following

acute coronary syndrome (ACS). The study aimed to provide a comprehensive understanding of the multifaceted factors influencing readmission, including socio-demographic characteristics, behavioral factors like smoking, and comorbidities such as diabetes and hypertension. The findings revealed that considering a wide range of variables, including cardiac-related complications, is essential for developing robust predictive models. This study is relevant to our project as we incorporate a similar variety of clinical and demographic features to improve the accuracy and effectiveness of our readmission prediction model.

However, most existing studies focus on small-scale datasets using traditional machine learning libraries; therefore, this research addresses the gap by implementing a scalable predictive framework using **PySpark** to handle hospital readmission data efficiently.

The implementation utilizes **Apache Spark** due to its distributed computing capabilities, which provide the necessary scalability to process large-scale healthcare datasets that exceed the limitations of traditional analytical tools.

TABLE I
SUMMARY OF STUDIES

Ref	Data	Target	Preprocess	Model	Results
[2]	Electronic health record (EHR) data combined with nursing assessments (e.g., ward severity, fall risk) for high-risk patients	Predict unplanned 30-day hospital readmission	Data preprocessing, integration of clinical, demographic, and nursing variables, and comparative analysis	Random Forest, CatBoost, and other machine learning models	Nursing and clinical features such as BMI, systolic blood pressure, and age were key predictors; early integration of nursing data improved identification of high-risk patients
[3]	Diabetic patient dataset	Predict hospital readmission	Data preprocessing and comparative analysis of patient records	Logistic Regression, Decision Tree, SVM, Naïve Bayes, and Neural Networks	SVM achieved the highest performance among the compared models
[4]	Medication-related and patient history data for older adults	Identify patients at higher risk of medication-related readmission	Analysis of medication usage, prior hospital visits, and chronic conditions	Risk assessment tool	Medication-related factors were found to significantly influence readmission risk
[5]	Electronic health record data from adult hospital admissions	Predict 30-day hospital readmission	Data preprocessing, clinical and demographic feature analysis, hyperparameter tuning, and internal validation	Logistic Regression, Random Forest, and LightGBM	Random Forest achieved the best performance and improved the identification of high-risk readmission patients
[6]	Findings from 24 previous studies on acute coronary syndrome readmission	Identify key factors affecting 30-day readmission	Literature analysis and comparison of previous studies	Various statistical and analytical approaches across reviewed studies	Readmission was influenced by socio-demographic factors, behavioral factors, comorbidities, and cardiac-related complications

III. EXPLORATORY DATA ANALYSIS

This phase employs the distributed computing framework of **PySpark** to conduct a systematic statistical investigation of the clinical indicators. The analysis focuses on validating data integrity, quantifying feature distributions, and uncovering underlying correlations related to hospital readmission, establishing a robust empirical foundation for the subsequent predictive modeling stages.

A. Data Quality and Preprocessing

The initial data quality check confirmed that the dataset contains no missing values or duplicate patient IDs, ensuring a robust foundation for analysis. For analytical purposes, the `blood_pressure` feature—originally formatted as 'systolic/diastolic'—was processed to extract the systolic blood pressure as a numeric feature, as it is a critical indicator of cardiovascular risk. Furthermore, the target variable `readmitted_30_days` was converted into a binary format (0 for No, 1 for Yes) to facilitate statistical computations.

B. Descriptive Statistics and Outlier Detection

Summary statistics reveal that the average patient age is approximately 53.8 years, with a mean BMI (Body Mass Index) of 28.9 and an average hospital stay of 5.5 days. To ensure data integrity, Outlier Detection was conducted using the Interquartile Range (IQR) method across numeric features, including age, BMI, cholesterol, and systolic blood pressure. The results indicated no significant outliers, suggesting that the clinical measurements fall within expected ranges.

C. Target Variable Distribution

Analyzing the distribution of the target variable is essential for understanding class balance. The data shows a significant imbalance, as the majority of patients were not readmitted within 30 days. This imbalance suggests that predictive models will require specific evaluation metrics, such as F1-score rather than simple accuracy.

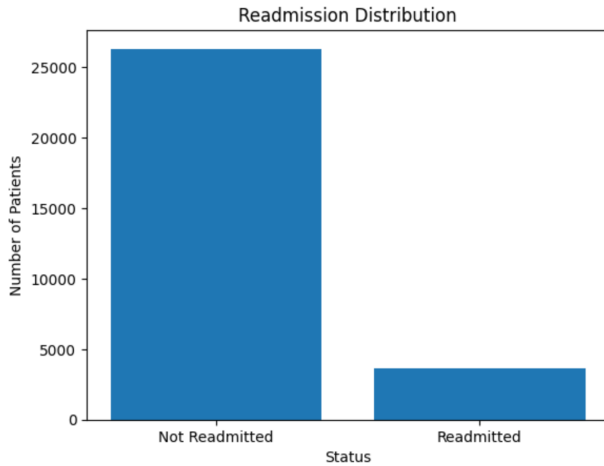


Fig. 1. Distribution of 30-day hospital readmission showing an imbalanced dataset.

D. Feature Correlation Analysis

A Correlation Heatmap was generated to examine the linear relationships between numerical variables. The analysis revealed that most correlation coefficients are close to zero, indicating that the features are largely independent of one another. This independence is beneficial for machine learning models as it reduces multicollinearity and ensures each feature provides unique information.

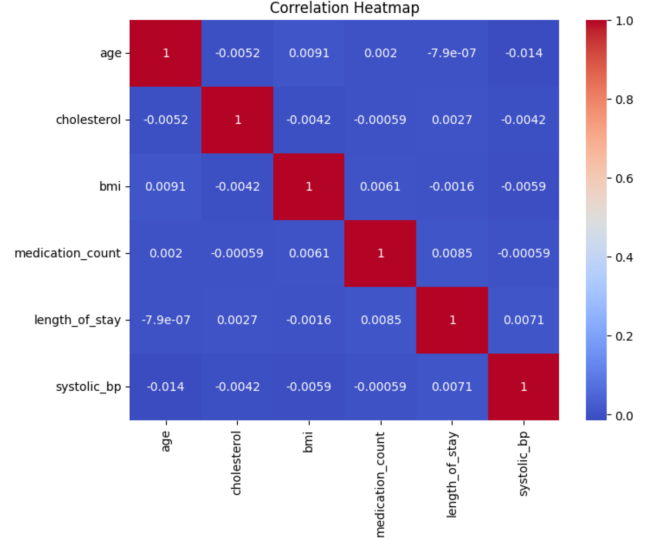


Fig. 2. Correlation heatmap of numerical patient features.

E. Impact of Chronic Conditions on Readmission

A key finding of the EDA involves the relationship between chronic conditions and readmission rates. It was observed that patients with diabetes only have the highest readmission risk (13.3%), followed by patients suffering from both diabetes and hypertension (13.1%), or only hypertension (12.9%), compared to 9.7% for patients with neither. This confirms that comorbidities significantly increase the likelihood of hospital readmission.

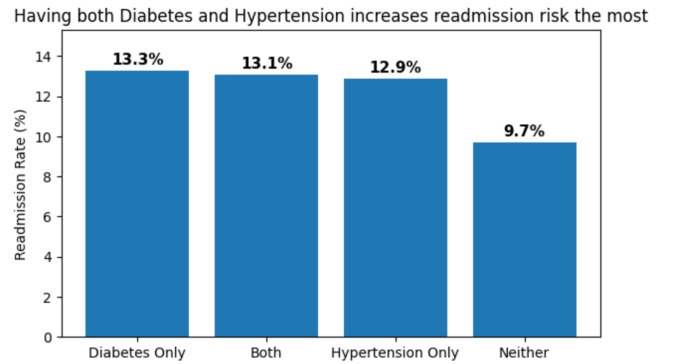


Fig. 3. Readmission rates based on Diabetes and Hypertension status.

F. Impact of Discharge Destination on Readmission

A key finding of the EDA involves the relationship between discharge destination and hospital readmission rates. It was observed that patients discharged to rehabilitation centers and nursing facilities have higher readmission rates compared to those discharged home. In particular, patients sent to rehabilitation centers show the highest readmission risk, followed by those transferred to nursing facilities, while patients discharged home have the lowest rate. This indicates that patients requiring additional post-discharge care are more likely to be readmitted, highlighting discharge destination as an important factor in predicting hospital readmission.

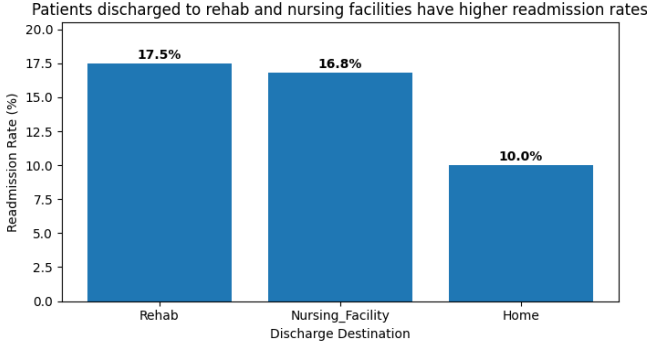


Fig. 4. Readmission Rate by Discharge Destination

IV. MODEL BUILDING

A. Machine Learning Models

- 1) **Logistic Regression:** The Logistic Regression model was implemented using PySpark MLlib to predict whether a patient would be readmitted within 30 days. A machine learning pipeline was constructed using StringIndexer, OneHotEncoder, and VectorAssembler to preprocess categorical and numerical features. The dataset was divided into training and testing sets, and the model was trained using the training data. Cross-validation and hyperparameter tuning were applied to improve model performance. The trained model was evaluated using Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC) to assess classification performance [7].
- 2) **Decision Tree:** A Decision Tree classification model was developed to classify patients based on their likelihood of hospital readmission. The pipeline included feature encoding and feature vector assembly before training the model. The Decision Tree model was trained using the training dataset and evaluated on the testing dataset. Performance metrics including Accuracy, Precision, Recall, F1-score, and AUC were used to evaluate the effectiveness of the model in predicting 30-day hospital readmission [9].
- 3) **Random Forest:** The Random Forest classification model was implemented using PySpark MLlib to improve prediction performance through ensemble learning. The

pipeline included categorical feature indexing, one-hot encoding, and vector assembly for feature preparation. The dataset was split into training and testing subsets, and the Random Forest model was trained using the training data. Evaluation metrics such as Accuracy, Precision, Recall, F1-score, and AUC were calculated to measure the prediction capability and generalization performance of the model. Feature importance analysis was also performed to identify the most influential factors affecting hospital readmission prediction [8].

B. Model Training and Testing using PySpark MLlib

1) **Data Splitting:** The dataset was divided into training and testing sets using a 70:30 split. The training set was used to train the machine learning models, while the testing set was used to evaluate model performance on unseen data. This split helps assess how well the models generalize to new patient records.

2) **Feature Processing:** Before training the models, the dataset was prepared for PySpark MLlib. Categorical variables such as gender, diabetes, hypertension, and discharge destination were converted into numerical representations using StringIndexer and OneHotEncoder. Numerical features such as age, cholesterol, BMI, medication count, length of stay, and systolic blood pressure were combined with the encoded categorical features using VectorAssembler. The final output was a unified feature vector used as input for the machine learning models.

3) **Evaluation Metrics:** The following metrics were used to evaluate the classification models:

Accuracy measures the proportion of correctly classified instances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Precision measures the correctness of positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

Recall measures the ability to identify all actual positive cases:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

F1-Score is the harmonic mean of Precision and Recall:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

AUC-ROC measures the model's ability to distinguish between readmitted and non-readmitted patients across all classification thresholds. A value closer to 1 indicates better model performance, while a value of 0.5 indicates random classification [10][11][12].

where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives.

V. EXPERIMENTS

The developed machine learning models were experimentally evaluated to assess their effectiveness in predicting 30-day hospital readmission using healthcare patient data. Multiple classification metrics were used to measure model performance, including Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC). Since the dataset contains class imbalance, the F1-score was considered one of the most important evaluation metrics because it provides a balanced measure between Precision and Recall.

A. Model Comparison

TABLE II

PERFORMANCE EVALUATION RESULTS OF MACHINE LEARNING MODELS

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	0.8764	0.7681	0.8764	0.8187	0.5884
Decision Tree	0.8751	0.7680	0.8751	0.8180	0.4349
Random Forest	0.8764	0.7681	0.8764	0.8187	0.5831

The experimental results demonstrated that Logistic Regression and Random Forest achieved the best overall performance among the developed models. Both models obtained the highest F1-score and AUC values, indicating stronger capability in distinguishing between readmitted and non-readmitted patients compared to the Decision Tree model. Furthermore, feature importance analysis using Random Forest revealed that cholesterol level, BMI, discharge destination, and age were among the most influential factors affecting hospital readmission prediction.

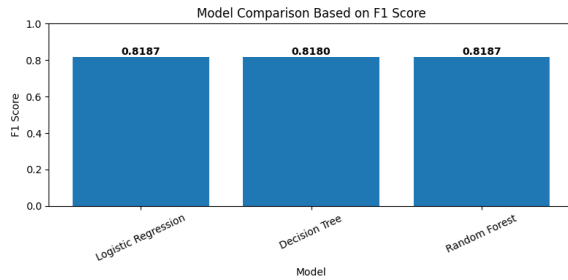


Fig. 5. Model Comparison Based on F1 Score

Figure 5 shows the comparison of machine learning models based on F1-score. Logistic Regression and Random Forest achieved the highest F1-score values, while Decision Tree showed slightly lower performance.

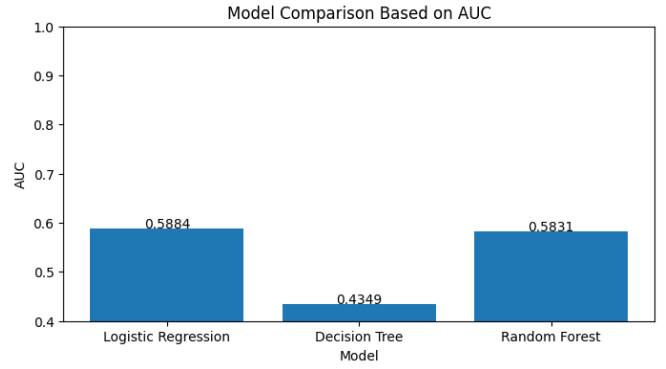


Fig. 6. Model Comparison Based on AUC

Figure 6 shows the comparison of machine learning models based on AUC values. Logistic Regression achieved the highest AUC score, followed closely by Random Forest, while Decision Tree recorded the lowest AUC value, indicating weaker classification capability.

VI. CONCLUSION

In this study, we successfully developed and evaluated a scalable, distributed machine learning framework using PySpark to predict 30-day hospital readmissions. By analyzing a healthcare dataset of 30,000 records, this research addressed the limitations of traditional analytical tools, demonstrating how Apache Spark's distributed architecture provides the necessary infrastructure for processing high-volume medical data efficiently. Our systematic exploratory data analysis confirmed that clinical and institutional factors are primary drivers of readmission. Specifically, chronic conditions were identified as major risk factors, with patients suffering from diabetes only (13.3%) and those with both diabetes and hypertension (13.1%) exhibiting the highest readmission rates. Furthermore, the study highlighted the impact of discharge paths, where patients sent to rehabilitation centers (17.5%) and nursing facilities were significantly more likely to return to the hospital compared to those discharged home. Experimental results showed that Logistic Regression and Random Forest models achieved the most robust performance, both delivering an identical F1-score of 0.8187, while Logistic Regression provided the optimal classification boundary with the highest AUC value of 0.5884. Ultimately, integrating these predictive pipelines into hospital systems directly supports Sustainable Development Goal 3 (SDG 3) by enhancing clinical decision-making, optimizing institutional resources, and improving the quality of patient care. Future work will explore the application of deep learning models and real-time streaming data to further enhance prediction precision.

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